

YOLO Dataset Configuration and File Structure Analysis

Introduction

YOLO is a real-time object detection and segmentation model that relies on structured datasets and configuration files. These configuration files help the model understand where data is located and how it is labeled.

Dataset YAML File

The dataset configuration file (e.g., coco128-seg.yaml) defines the structure and metadata of the dataset. It includes:

- **Path:** Specifies the root directory of the dataset
- **Train/Validation Paths:** Define locations of training and validation images
- **Class Names:** Maps class indices to object names

Example:

- 0 → person
- 1 → bicycle
- 2 → car

This mapping allows the model to interpret predictions correctly.

Folder Structure

A typical YOLO dataset is organized as follows:

- images/train – Training images
- images/val – Validation images
- labels/train – Corresponding label files for training
- labels/val – Corresponding label files for validation

Each image has a corresponding label file with the same name.

Label File Format

Bounding Box (Detection)

Each object is represented as:

- class_id x_center y_center width height

All values are normalized between 0 and 1.

Example:

0 0.5 0.5 0.3 0.4

Segmentation Format

In segmentation tasks, objects are represented using polygon coordinates:

- class_id x1 y1 x2 y2 x3 y3 ...

This allows pixel-level classification of objects.

Conclusion

The YOLO dataset configuration and file structure provide a standardized way to organize data for training and evaluation. The YAML file defines dataset metadata, while label files provide annotations for object detection and segmentation. This structured approach enables efficient model training and accurate predictions.

References

1. Ultralytics YAML Configuration Guide

<https://www.ultralytics.com/glossary/yaml>

2. YOLO Dataset Format Documentation

https://open-edge-platform.github.io/datumaro/stable/docs/data-formats/formats/yolo_ultralytics.html

3. Ultralytics YOLO Dataset Structure (CVAT Docs)

https://docs.cvat.ai/docs/dataset_management/formats/format-yolo-ultralytics/